RiskPulse AI

Explainable risk intelligence

From unstructured information to auditable portfolio impact

AI / NLP risk engine + deterministic stress testing

Candidate: Amit Kumar

College / Campus: IIIT Raichur

S&P Global & Crisil Campus Hackathon



SIGNAL
to IMPACT

Problem & approach

The business problem

Risk teams receive fragmented, unstructured information. A headline or social post may indicate credit, geopolitical, regulatory, or market stress, but raw text does not explain which portfolio positions are exposed or how severe the impact could be.

- Information is high-volume and difficult to prioritize.
- Entity names, event types, and exposure links are easy to miss.
- Naive stress tests are disconnected from their triggering evidence.

The RiskPulse AI approach

A traceable pipeline links source records to risk signals and then to deterministic, auditable portfolio stress results.

01 **Ingest**
RSS and historical social CSV

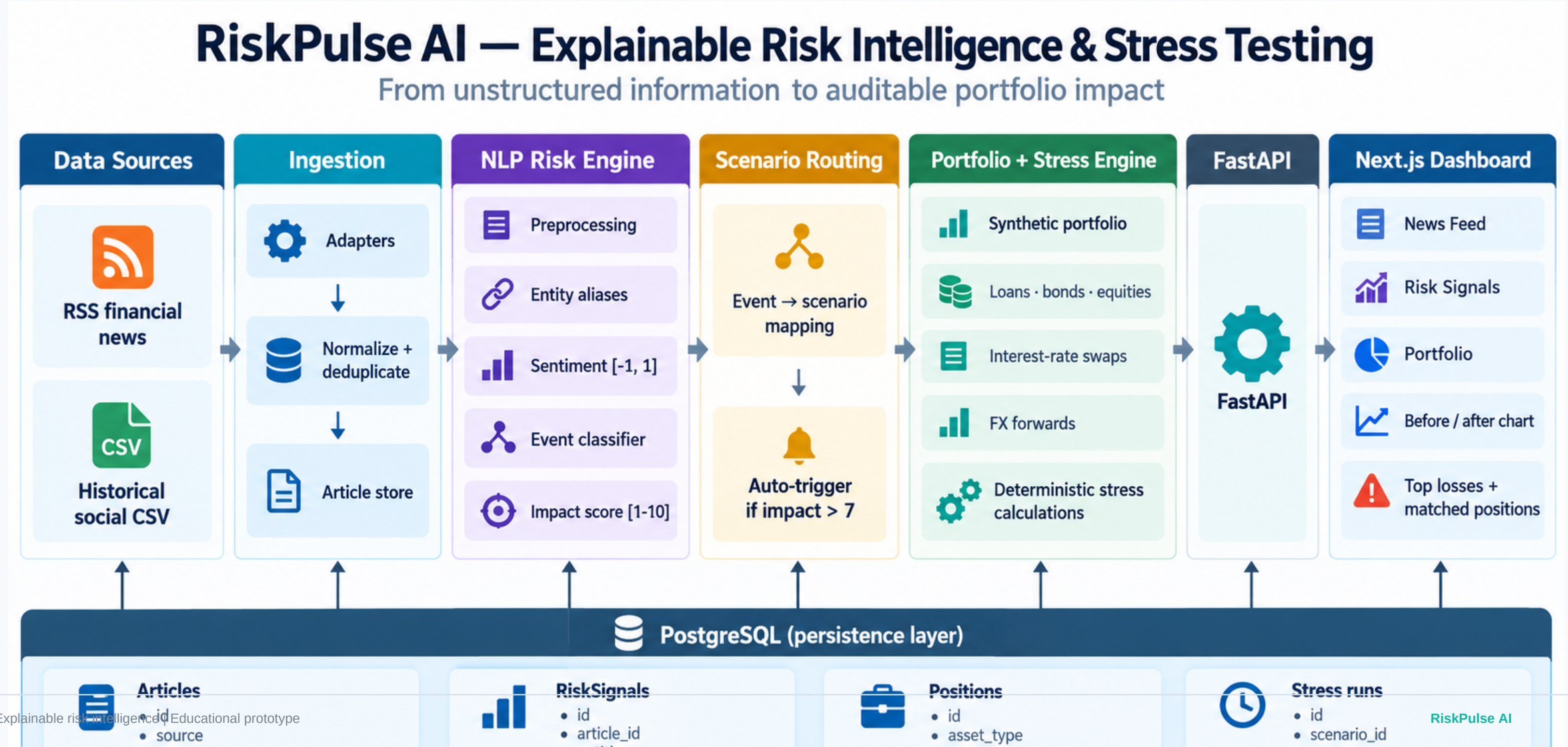
02 **Understand**
Entities, sentiment, event type

03 **Score**
Impact 1-10 with factor explanation

04 **Stress**
Scenario routing and position P&L

System design & data flow

A source record becomes a persisted, explainable stress result.



Implementation highlights

Ingestion

RSS, social CSV replay, optional Reddit, normalized timestamps, content hashes, source-level status, bounded retries.

NLP risk engine

Alias-driven entities, heuristic default, optional FinBERT, event classifier, sentiment [-1, 1], impact [1-10].

Stress engine

One event-to-scenario mapping, targeted eligibility, corporate credit, equities, bonds, swaps, and FX forwards.

Auditability

PostgreSQL persistence, Alembic migrations, input snapshots, calculation details, reconciliation, and triggering article links.

Technology choices

- FastAPI + SQLAlchemy + PostgreSQL: clear service boundaries and durable audit records.
- Next.js + TypeScript + Recharts: fast analyst-facing views for feed, signals, portfolio, and stress results.
- Deterministic scoring and scenario valuation: repeatable demo behavior and testable assumptions.

Design principle Every material result should answer: what happened, why it mattered, and which positions were affected?

Key results

82
backend tests passing

2
data sources in the demo

20
seeded portfolio positions

7
default auto-trigger threshold

1.0000
smoke accuracy / macro-F1

Naive flow vs. RiskPulse AI

Naive feed

- Raw articles without exposure linkage
- No consistent score range or rationale
- Manual scenario selection
- No position-level reconciliation

RiskPulse AI

- Named issuer and ticker aliases map signals to positions
- Sentiment, event weight, and entity exposure explain impact
- Impact above 7 automatically routes a scenario
- Before/after P&L, top losses, matched positions, and reconciliation

The smoke metrics are deliberately qualified: the local PhraseBank fixture has three rows and is not a full benchmark.

Domain impact

A practical analyst workflow

01 Detect → **02 Explain** → **03 Prioritize** → **04 Act**

A new article or historical social record enters the same normalized pipeline.

Entity, event type, sentiment, and impact factors are visible instead of hidden in a black box.

A threshold identifies material signals and selects the matching stress scenario.

The analyst sees portfolio value change, top losses, and matched exposures for review.

Business value

Creates a common language between information, risk signals, exposure, and scenario results. This supports faster triage and more defensible analyst conversations.

Demo value

The workflow is deterministic, inspectable, and easy to replay. A reviewer can move from a triggering article to the exact positions included in the stress result.

Limitations & next steps

Current limitations

This is a case-study prototype. The output is designed to demonstrate traceability and scenario mechanics, not to represent a calibrated production risk model.

- Portfolio holdings are synthetic; transaction categories are proxies, not proof of real exposures.
- Stress shocks are illustrative assumptions and are not forecasts or investment advice.
- Historical social replay lacks live delivery order, deleted-post behavior, engagement data, and return labels.
- Heuristic sentiment is the safe default; FinBERT requires model download and legal review.

Next steps

The foundation can evolve toward a governed risk workflow with stronger data and model controls.

- Replace synthetic holdings with permissioned portfolio and reference-data integrations.
- Calibrate shocks and validate results against approved market and credit methodologies.
- Add authentication, role-based access, secrets management, observability, and deployment controls.
- Expand evaluation beyond the three-row smoke fixture and monitor model drift.

Demo sequence Ingest demo articles -> process NLP -> open Stress Testing -> inspect triggering article and matched positions